



Cambridge International AS & A Level

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MATHEMATICS

9709/41

Paper 4 Mechanics

May/June 2026

1 hour 15 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

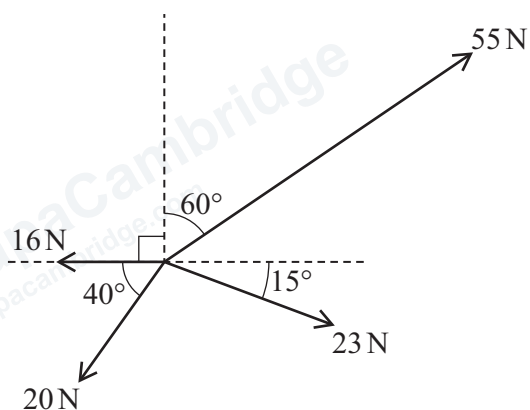
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages.



1



Coplanar forces of magnitudes 55 N, 23 N, 20 N and 16 N act at a point in the directions shown in the diagram.

Find the magnitude and direction of the resultant force.

[6]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines spaced evenly down the page. Diagonal dotted lines cross these horizontal lines at approximately a 45-degree angle, creating a series of slanted rectangular boxes across the entire surface. This design is typical for teaching handwriting or basic geometry to young students. There are no margins, text, or other markings on the page.

- 2 A car of mass 1200 kg is moving on a straight horizontal road. The resistance to the motion of the car has a constant magnitude of 600 N and the car's engine is working at a rate of 27 kW.

(a) Find the acceleration of the car when its speed is 18 m s^{-1} . [3]

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The car now moves up a straight hill which is inclined at an angle θ to the horizontal. The resistance to the motion of the car has now changed to a constant magnitude of 650 N. The car moves up the hill at a constant speed of 12 m s^{-1} with the car's engine working at a rate of 22 kW.

(b) Find the value of θ . [3]

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- 3** A particle P travels in a straight line, starting from a point O . At time t seconds the displacement of P from O is s metres, where $s = 4t^3 - 39t^2 + 120t$.

(a) Find the values of t for which P is instantaneously at rest. [3]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dotted lines, each set separated by a larger gap. A diagonal dashed line runs from the top-left towards the bottom-right, bisecting the page. The background is white, and there are faint, repeating watermarks of the text "PapaCambridge.com" across the page.

(b) Find the total distance travelled by P in the interval $1 \leq t \leq 4$.

[2]

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(c) Find the magnitude of the acceleration of P when $t = 3$.

[3]

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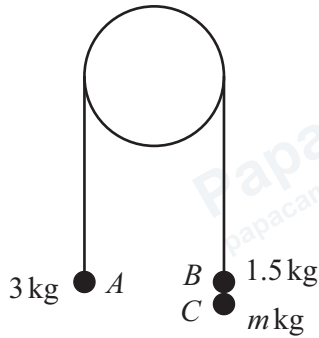
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Three particles A , B and C have masses 3 kg , 1.5 kg and $m\text{ kg}$ respectively, where $m > 1.5$. Particles A and B are attached to the ends of a light inextensible string that passes over a smooth fixed pulley.

Particle C is attached to particle B . The system is held at rest with the string taut and the hanging parts of the string vertical (see diagram).

The system is released from rest and while the three particles are all attached as described, the tension in the string is 37.5 N.

- (a)** Find the value of m .

[4]

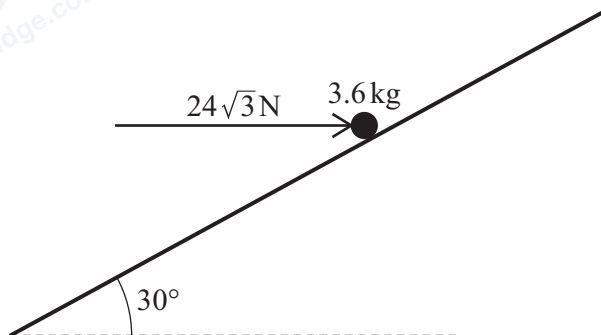
[illegible]



At the instant when A has moved a distance 0.8 m upwards from its initial position, C separates from B and falls away. In the subsequent motion, A and B continue to move and A does not reach the pulley.

- (b)** Find the total time from when the system is released until A first comes to instantaneous rest. [6]

[illegible]



A particle of mass 3.6 kg is placed on a rough plane inclined at an angle of 30° to the horizontal. The particle is held at rest on the plane by a horizontal force of magnitude $24\sqrt{3}\text{ N}$. The force acts in a vertical plane containing a line of greatest slope of the inclined plane (see diagram). The particle is on the point of moving up the plane.

- (a) Show that the coefficient of friction between the particle and the plane is $\frac{1}{5}\sqrt{3}$. [5]

[illegible]



The force of magnitude $24\sqrt{3}\text{ N}$ is now removed, and the particle moves from rest down the plane.

- (b)** Find the speed of the particle after it has travelled 1.5 m down the plane. [5]

[illegible]



- 6 Three small smooth spheres A , B and C of equal radii and masses m kg, 2 kg and 4 kg respectively, lie at rest in that order in a straight line on a smooth horizontal plane. Sphere A is projected directly towards B with speed $2m$ m s⁻¹ and B is projected directly towards A with speed $(m + 2)$ m s⁻¹. After the spheres collide, both A and B move in the same direction as A 's original motion with speeds 2 m s⁻¹ and 3.25 m s⁻¹ respectively.

(a) Find the value of m .

[4]

[illegible]



(b) Determine if there is a further collision between A and B .

[6]

This image shows a full page of blank handwriting practice paper. It features multiple rows of horizontal lines. Each row consists of a solid top line, a dashed midline, and a solid bottom line, providing a guide for letter height and placement. The background is white, and the lines are light gray. Faint, diagonal watermarks reading "PapaCambridge" and "papacambridge.com" are repeated across the entire page.

[illegible]

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